

# Weekly question week 5

## Grade XI Physics

### Assignment Submission Date: Bhadra 1

- a. If  $\vec{A}$  and  $\vec{B}$  are non-zero vectors, is it possible  $\vec{A} \times \vec{B}$  and  $\vec{A} \cdot \vec{B}$  both to be zero at the same time? Explain.
- b. What do you mean by resolution of a vector?
- c. Discuss in detail, the resolution of a vector in two dimensions.
- d. Explain the geometrical interpretation of scalar product of two vectors.
- e. Vectors;  $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$  and  $\vec{B} = -3\hat{j} + 6\hat{k}$ . Find the vector  $\vec{C}$ , such that  $2\vec{A} + 7\vec{B} + 4\vec{C} = 0$ .
- f. Find the scalar product of the two vectors:  $\vec{A} = 4.00\hat{i} + 3.00\hat{j}$  and  $\vec{B} = 5.00\hat{i} + 2.00\hat{j}$ . Also, find the angle between them.
- g. a. Two spheres of same radius are made up of same material but one is hollow and the other is solid. Will their overall expansion be same or different if they are heated from  $20^\circ\text{C}$  to  $30^\circ\text{C}$ ? Explain?
- h. A Steel rod and a brass rod when measured at  $0^\circ\text{C}$  are 50.1 cm and 50cm respectively. At what temperature will their length become equal and what will be their common length? Given,  $\alpha_{\text{steel}} = 12 \times 10^{-6}/^\circ\text{C}$  and  $\alpha_{\text{brass}} = 18 \times 10^{-6}/^\circ\text{C}$ .
- i. Does the density of a substance depend on temperature? Explain.
- j. Derive an expression showing the variation of density of a substance with the change in temperature.
- k. The coefficient of real expansion of a substance is  $1.6 \times 10^{-4}/^\circ\text{C}$ . If the density of the substance is 10 g/cc at  $0^\circ\text{C}$ , find its density at  $673\text{K}$ .
- l. On heating a glass block of 10,000  $\text{cm}^3$ , from  $25^\circ\text{C}$  to  $40^\circ\text{C}$ , its volume increases by  $4\text{cm}^3$ . Calculate coefficient of linear expansion of the glass.
- m. The difference between the lengths of a certain brass rod and that of a steel rod is claimed to be constant at all temperatures. Is this possible? Explain.
- n. Describe the experiment method for the determination of linear expansivity of a material in the form of a rod.
- o. A glass rod rubbed with silk is found to have acquired a positive charge of  $6.4 \times 10^{-7}\text{C}$ . Does the glass rod lose any mass in this process? If so, estimate the loss of mass.
- p. State and explain Coulomb's law in electrostatics.
- q. Point Charges having values  $+5\mu\text{C}$ ,  $+10\mu\text{C}$ , and  $-10\mu\text{C}$  are placed in corners A, B and C of an equilateral triangle ABC having each side equal to 5 cm. Find the resultant force on the charge at A.
- r. State and explain the principle of superposition of Coulomb's force.
- s. Why coulomb's law is applicable to stationary charges only?
- t. Two point charges of  $10\mu\text{C}$  and  $-90\mu\text{C}$  are separated by a distance of 18 cm. Find the point along the line joining them at which electric field intensity is zero.